Load Transportation Robot Case Study

The system consists of two mobile robots where each robot is a differential drive wheeled mobile robot, capable of traversing in an environment, which is surrounded by four walls. The task of the robots is to find a passage and exiting through the passage while holding one large object on top of them. Therefore, the main goal of these mobile robots is to transfer one large object through an unknown passage. Leader-followers strategy has been used where one mobile robot will be a leader while the other will be a follower.

The robot consists of a body with 300 mm diameter and two pair of wheels. The robot's body has four layers including a ground layer. The ground layer consists of a pair of drive wheels and a pair of castor wheels. Each drive wheels are attached to DC motors to provide movement. While the castor wheels are for stabilization. This layer is about 150 mm high. The height of the first layer is about 170 mm. The battery and driver for DC motors are place at this level. Infrared (IR) proximity sensor for obstacle avoidance is also located at this level. The second layer is about 220 mm high. It places main processor, Radio Frequency (RF) module for inter-robot communication, IR proximity sensor for wall detection and IR transceiver for landmark recognition. The third layer is for the supporting base. The mobile robot component configuration is shown in Figure 1. Figure 2 illustrates a top view of the robot.

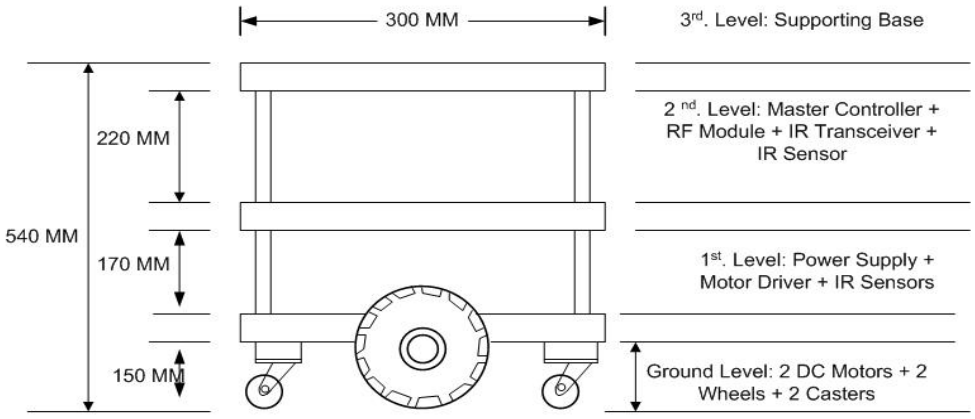


Figure 1 Components configuration of the robot

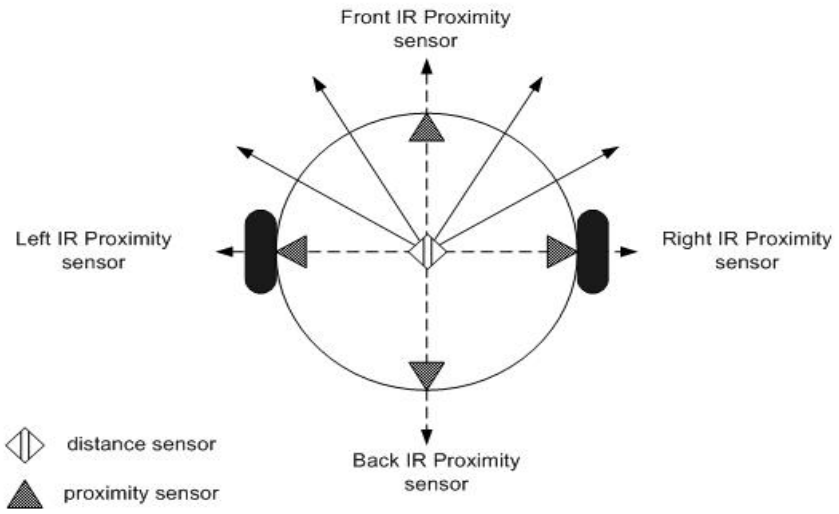


Figure 2 Top view of the robot

The software has to run on embedded processor. The main processor in the controller is based on Motorola68HC11 microcontroller with 32K RAM, Parallel I/O, Serial I/O, ADC/DAC, Switches on/stop and Status Liquid Crystal Display (LCD). The embedded controller can be represented in a block diagram form as in Figure 8.30. The main function of the controller is to move the wheels of the robot until the robot achieve its goal. In order to move the robot and achieve the goal, the software needs to control the motor at each robot drive wheels, monitor the environments, navigate the robot and cooperate with other robot. The controller monitors its environment using four IR proximity sensors and two IR distance sensors. The load detection is done via one push button. The controller also communicates with other robot via a Radio Frequency (RF) transceiver. The controller also communicates with human through LCD and a switch.

The embedded controller monitors its environment using some sensors. The environment must be monitored, typically every half a second to detect the presence of obstacle and wall using the IR proximity sensors during the movement of the robot. The speed of the motor is sensed using

the encoders and fed back to the embedded controller. The computation of the control signal typically, must be completed within 50 milliseconds to ensure the correct speed of the robot.

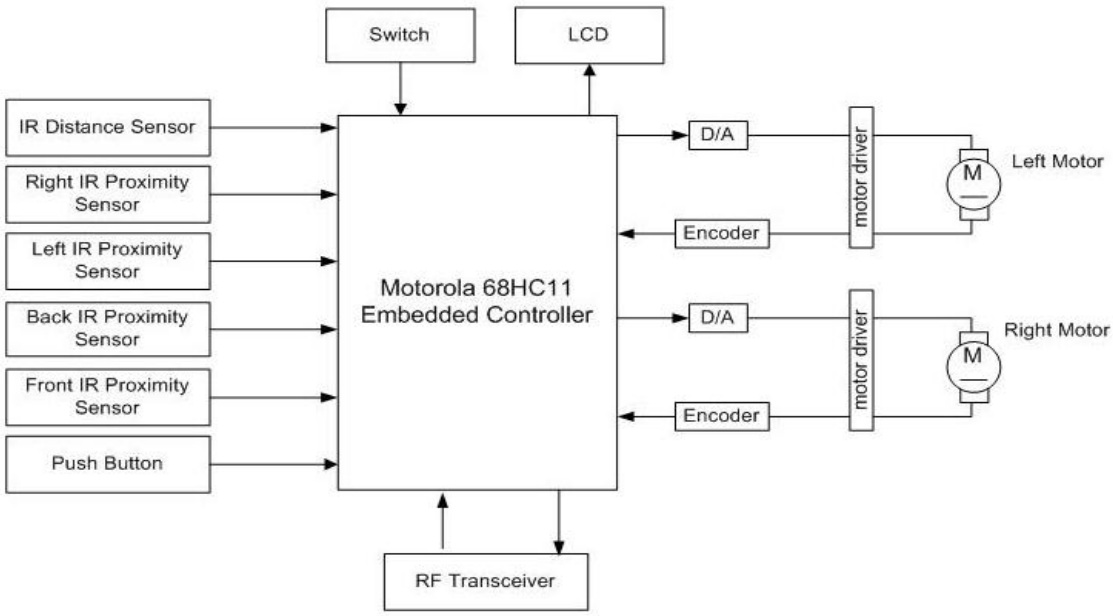


Figure 3 Block Diagram of the UTM intelligent AMR controller